

Neural Simulation-Based Inference for $gg \rightarrow (h^* \rightarrow) ZZ \rightarrow 4\ell$ Using EveNet Foundation Model



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Introduction + Context

- The Large Hadron Collider (LHC) smashes protons at near light-speed, letting us probe the fundamental building blocks of nature and test the Standard Model
- The Higgs boson gives the fundamental particles their mass. We measure how strongly it interacts via the signal strength μ . If $\mu = 1$, theory holds. If $\mu \neq 1$, could be a sign of new physics.
- We study $gg \rightarrow h^* \rightarrow ZZ \rightarrow 4\ell$: a clean, distinctive signal, but it is 20 $\times$ smaller than a background process that looks identical, $gg \rightarrow ZZ \rightarrow 4\ell$
- Traditional methods use only 1-2 features per event and discard the rest, throwing away most of the information in each collision

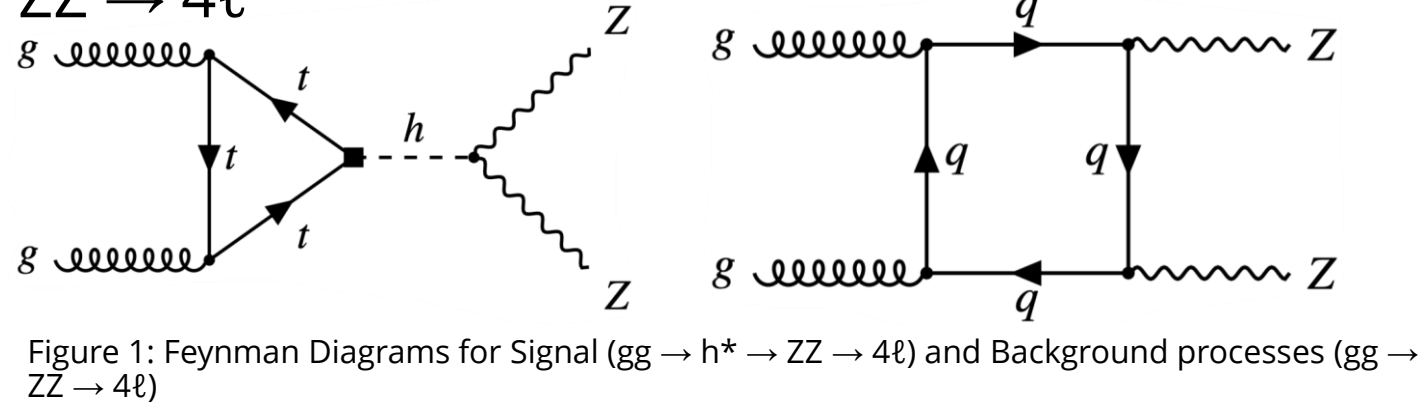
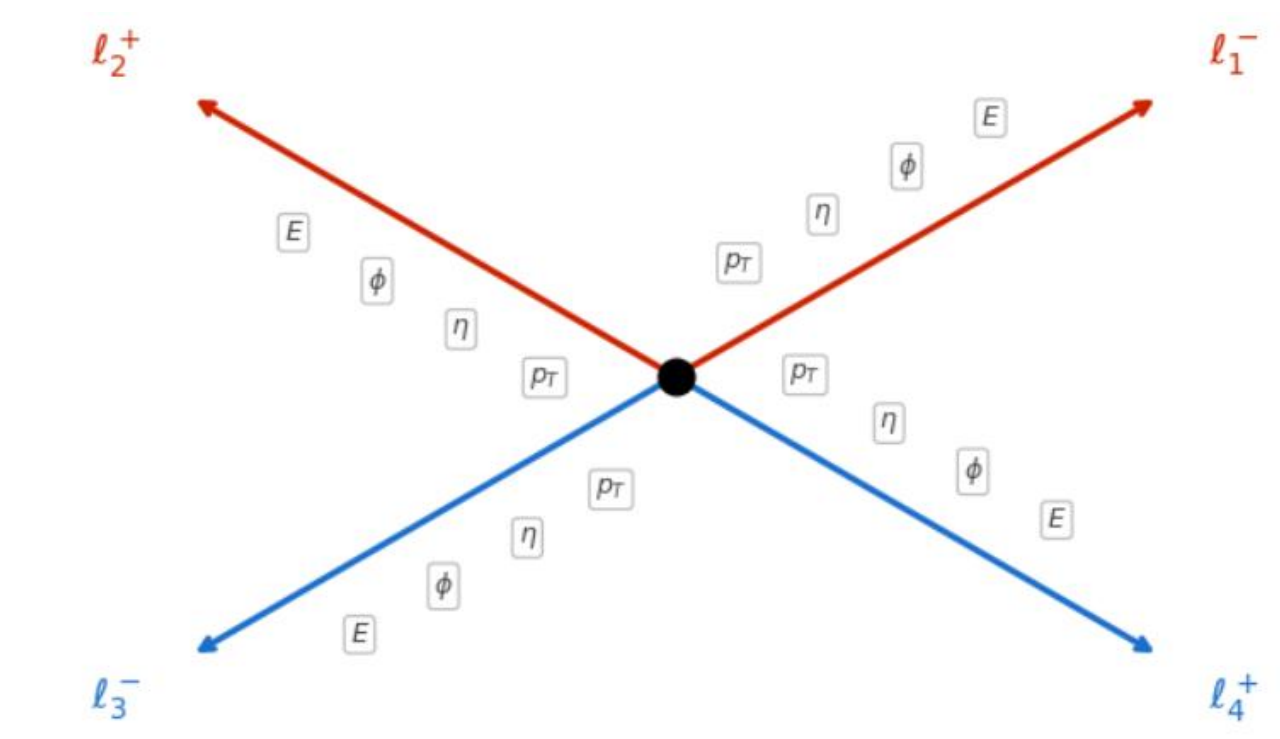


Figure 1: Feynman Diagrams for Signal ($gg \rightarrow h^* \rightarrow ZZ \rightarrow 4\ell$) and Background processes ($gg \rightarrow ZZ \rightarrow 4\ell$)

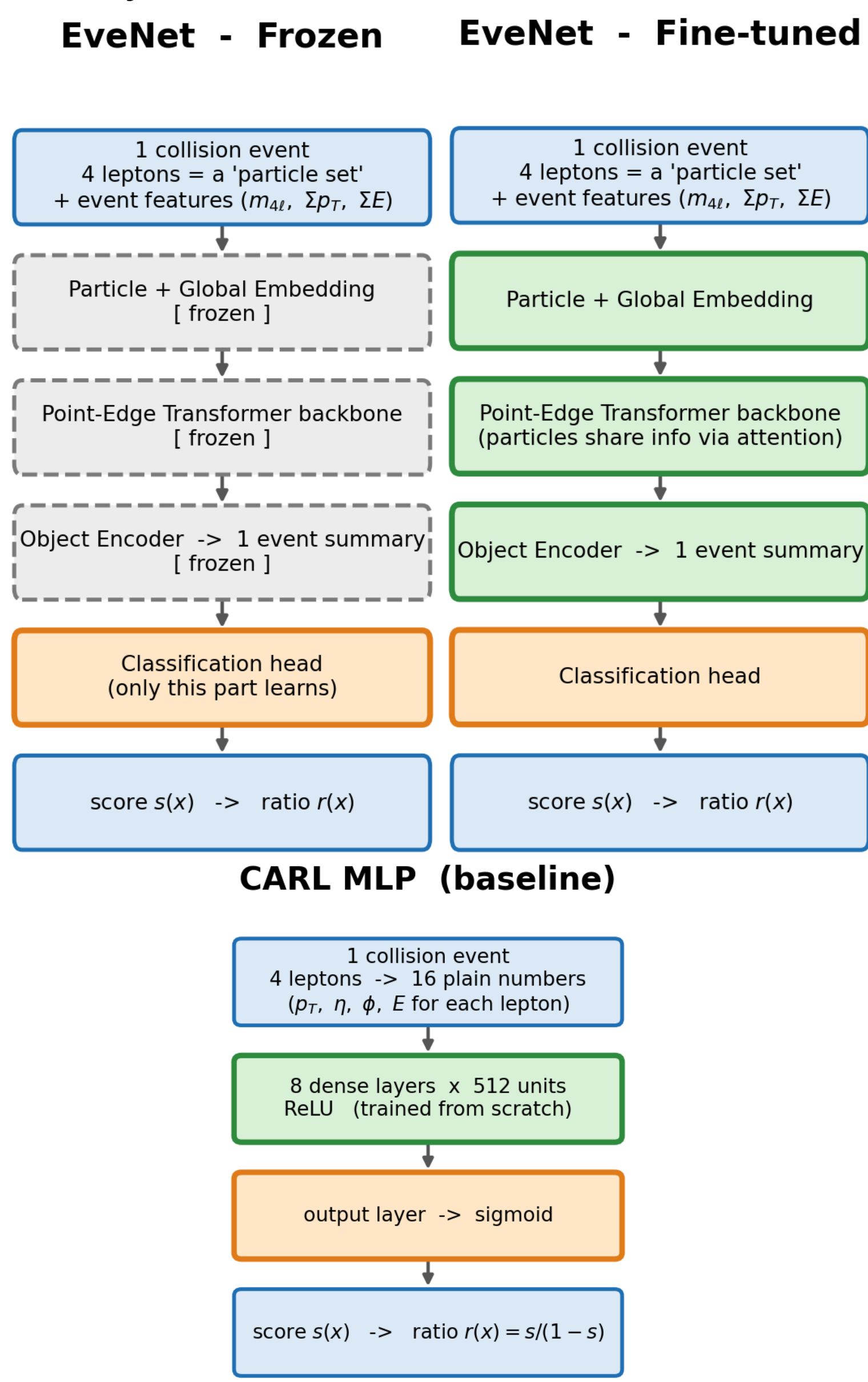


NSBI

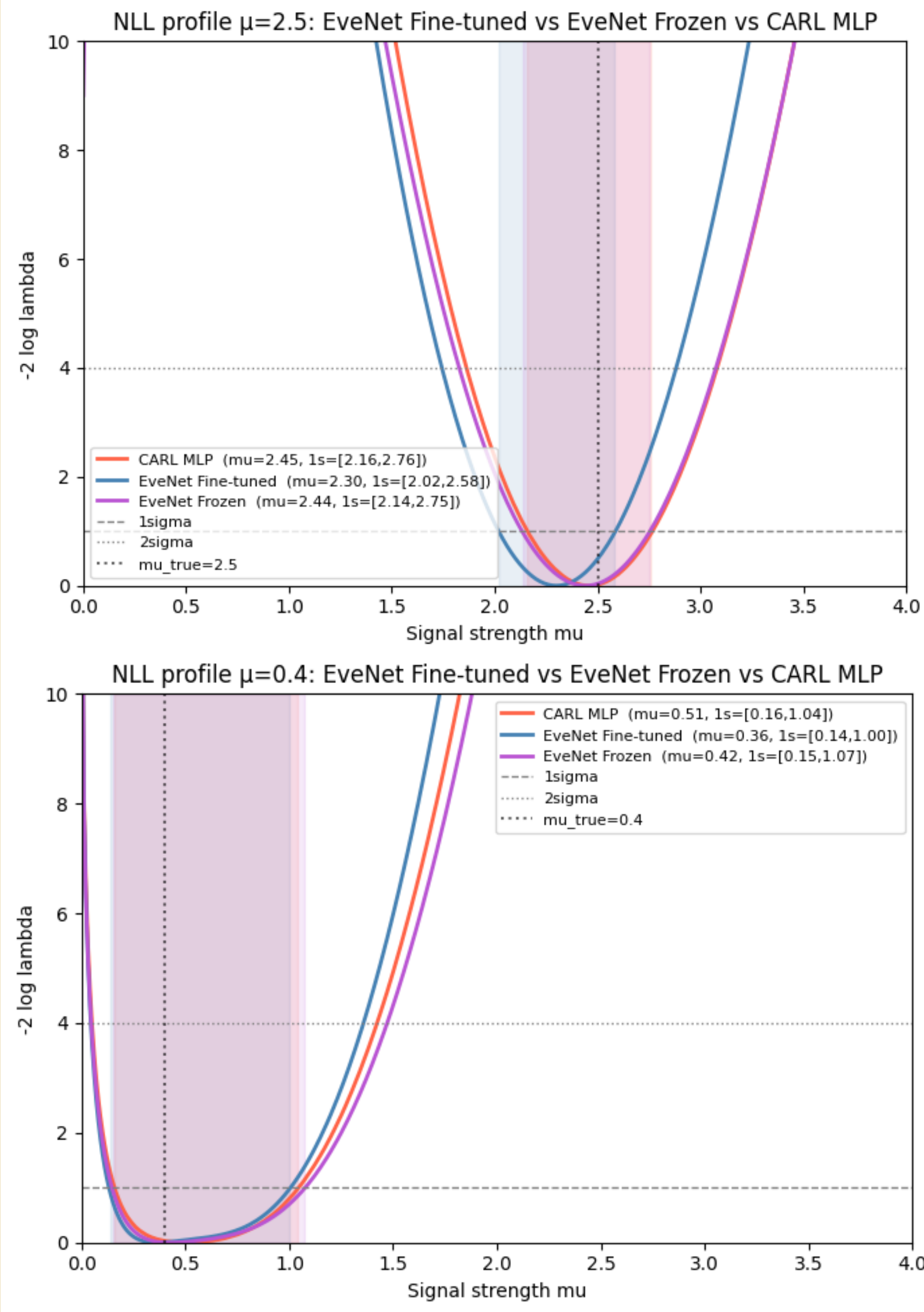
- Instead of picking, we train a neural network using all features, extracting the most information
- Learns to distinguish between signal and background using simulated data, using the output of the NN to calculate the likelihood ratio $r(x)$
- Using the likelihood ratio, all values of μ are scanned over, creating a map of the most likely μ values

Goal and Methodology

- EveNet is a foundation model for particle physics, a large transformer pretrained on simulated collision data to learn general representations of particle events.
- Goal is to implement NSBI into EveNet's classification head and test EveNet against traditional MLP (CARL)
- We test two adaptation strategies: freezing the backbone (Frozen) and fully fine-tuning all layers (Fine-tuned).



Results



- At high signal strength ($\mu = 2.5$), all three models recover the true value within one standard deviation
- At low signal strength ($\mu = 0.4$), all three models again recover the true value, each within one standard deviation
- Both EveNet strategies (Frozen and Fine-tuned) closely track the CARL MLP baseline at both signal strengths

Conclusion & Future Work

- NSBI successfully implemented into EveNet Foundation Model
- At both $\mu = 2.5$ and $\mu = 0.4$, all three models recover the signal strength within one standard deviation
- EveNet Frozen performs similarly to CARL MLP, suggesting that pre-trained representations transfer effectively to NSBI tasks
- EveNet Fine-tuned also recovers the true value at both $\mu = 2.5$ and $\mu = 0.4$, matching the CARL baseline
- Future work: extend to more μ values, larger event samples, additional fine-tuning strategies, and EveNet from Scratch

References

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